Muhammad Irfan (M.Phil Zoology & M.Ed) **03036601120**

FAST SCIENCE ACADEMY

Name: ..... Test: (Biology) Class: 12th

Date: .... /.... /..... Max Marks (85) Chapter: 15-19 Time: 2 Hour

Question No.1 Encircle the correct answer **17**

1. The fishes which drink large amount of sea water and excrete concentrated urine are:
a Hag fishes **b** Cartilaginous fishes **c** Bony fishes **d** Freshwater fishes

2. Which of the following is not synthesized in liver?
a Prothrombin **b** Urea **c** Ammonia **d** Urine

3. Most land mammals respond to cold by raising their:
a Skin **b** Fur **c** Spines **d** Bristles

4. Metabolism of purines and pyrimidines produce significant amount of:
a Creatine **b** Creatinine **c** Trimethylamine oxide **d** Xanthinae

5. The number of thoracic vertebrae is:
a 7 **b** 12 **c** 33 **d** 5

6. The protein that binds to Calcium in muscle contraction is:
a Actin **b** Myosin **c** Tropomyosin **d** Troponin

7. The joints that are held together by short fibers embedded in connective tissue are:
a Fibrous joints **b** Cartilaginous joints **c** Synovial joints **d** Hinge joints

8. Which neuron have long axons?
a Sensory **b** Motor **c** Associative **d** All are same

9. The cell membrane of neuron is permeable only for:
a Na^+ **b** K^+ **c** Ca^+ **d** None

10. In human relay centre is located in:
a Fore Brain **b** Mid brain **c** Hind Brain **d** Hypothalamus

11. Sperms are formed in:
a Vas deferns **b** Collecting duct **c** Epididymis **d** Seminiferous tubule

12. From which month the embryo of human is called as fetus?
a 3rd **b** 5th **c** 7th **d** 9th

13. Fusion of egg and sperm takes place in:
a Ovary **b** Uterus **c** Uterine tube **d** Cervix

14. Movement and rearrangement of cells in embryo is called:
a Cleavage **b** Morulation **c** Gastrulation **d** Neurulation

15. Neural plate is formed from:
a Ectoderm **b** Mesoderm **c** Endoderm **d** Notochord

16. Plants have growth pattern called:
a Open growth **b** Closed growth **c** Meristem **d** Growing point

17. Grey vegetal cytoplasm gives rise to:
a Muscles **b** Gut **c** Notochord **d** Neural tube

Question No.2 Short Questions **44**

1. Differentiate between osmoconformers and osmoregulators with the help of examples
2. Define xerophytes? How they are adapted to their environment?
3. State two functions of liver in relation to homeostasis?
4. Define and draw urea cycle?
5. Differentiate between excretophore and ebony.
6. Differentiate between hyaline fibrocartilage.
7. Differentiate between epicuticle and procuticle.
8. What is all or none principle?
9. What is sciatica? Write its causes.
10. What are commercial applications of Ethene?
11. What are neurotransmitters? Give their various types.
12. Write a note on Alzehmer's Disease.
13. Define acromegaly. Give its causes.
14. What is leutinizing hormone? Give its role.
15. What is apical dominance? Give its causes.
16. What are neoblasts?
17. Differentiate between epiblast and hypoblast.
18. Enlist the key events in animal development.
19. What is corpus callosum?
20. What is moulting?
21. What is T-system?
22. What are pyrogens?

Question No.3

A) Describe the mechanism of thermoregulation in animals. **4**
B) What is bone fracture? Explain repair process of a simple fracture. **4**

Question No.4

A) Describe the role of adrenal gland **4**
B) Write a note on birth. **4**

Question No.5

A) Describe sliding filament model of muscle contraction**.** **4**
B) Explain role of nucleus in development. **4**